

On the Constructive Dimension Spectrum of Polynomials

Prajval Koul & Satyadev Nandakumar

(Department of Computer Science and Engineering
Indian Institute of Technology Kanpur)

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- ② Main Theorem 1
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- ④ Future Scope
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- Algorithmic information of geometric objects.
- Effective Hausdorff dimension of points on polynomials.
- Dimension spectrum of polynomials.

Kolmogorov Complexity

For a pair of strings $u, v \in \{0, 1\}^*$, the Kolmogorov complexity of u given v is defined as

$$K(u|v) = \min_{\pi} \{\ell(\pi) : U(\pi, v) = u\}$$

where U is the universal (prefix) Turing machine.

The Kolmogorov complexity of a string $u \in \{0, 1\}^*$ is defined as

$$K(u) = K(u|\lambda)$$

where λ is the empty string.

Kolmogorov Complexity of reals (Lutz, Mayordomo, 2008)

For a given $x \in \mathbb{R}$, its Kolmogorov complexity at a precision $r \in \mathbb{N}$ is defined as

$$K_r(x) = \min \{ K(q) : q \in \mathbb{Q} \cap B(x, 2^{-r}) \}.$$

Effective Hausdorff Dimension (Lutz, Mayordomo, 2008)

For a given $x \in \mathbb{R}$, its effective Hausdorff dimension is defined as

$$\dim(x) = \liminf_{r \rightarrow \infty} \frac{K_r(x)}{r}.$$

Conditional Kolmogorov Complexity of reals (Lutz, Mayordomo, 2008)

For an $x \in \mathbb{R}$, its conditional Kolmogorov complexity at a precision $r \in \mathbb{N}$ given a $q \in \mathbb{Q}$ is defined as

$$K_r(x|q) = \min \{K(p|q) : p \in \mathbb{Q} \cap B(x, 2^{-r})\}.$$

The conditional Kolmogorov complexity of $x \in \mathbb{R}$ at precision $r \in \mathbb{N}$ given a $y \in \mathbb{R}$ at precision $s \in \mathbb{N}$ is define as

$$K_{r,s}(x|y) = \max \{K_r(x|q) : q \in \mathbb{Q} \cap B(y, 2^{-s})\}.$$

For $r = s$, $K_{r,s}(x|y)$ is denoted by $K_r(x|y)$.

Effective Hausdorff Dimension (Lutz, Mayordomo, 2008)

The effective Hausdorff dimension of $x \in \mathbb{R}$ given $y \in \mathbb{R}$ is defined as

$$\dim(x|y) = \liminf_{r \rightarrow \infty} \frac{K_r(x|y)}{r}.$$

Dimension Spectrum (Lutz, Mayordomo, 2020)

For a set $E \subseteq \mathbb{R}^2$, its dimension spectrum is defined as

$$sp(E) = \{dim(x, y) : (x, y) \in E\}.$$

Theorem (Lutz, Stull, 2020)

For $a, b \in \mathbb{R}$ (the slope-intercept pair representing a line) and $x \in \mathbb{R}$,

$$\dim(x, ax + b) = \dim(x|a, b) + \min \left\{ \dim(a, b), \dim^{a,b}(x) \right\}.$$

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Theorem (Stull, 2022 (ICALP), 2025 (J. LMS))

Let $a, b \in \mathbb{R}$ be the slope-intercept pair representing a line. Then, for every $s \in [0, 1]$, there is a point $x \in \mathbb{R}$ such that

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Open problem: Does an analogous result hold for the case of polynomials as well?

Theorem 1 (K., Nandakumar, 2026)

For $(a_0, a_1, \dots, a_d) \in \mathbb{R}^{d+1}$ and $x \in \mathbb{R}$, we have

$$\dim \left(x, \sum_{i=0}^d a_i x^i \right) \leq \dim(x|a_0, \dots, a_d) + \min \{ \dim(a_0, \dots, a_d), \dim^{a_0, \dots, a_d}(x) \}.$$

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Thus, every polynomial has at least 2 points with different effective Hausdorff dimension.

Proof:

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(This is done by showing that the r -bit approximation of $(x, \sum_{i=0}^d a_i x^i)$ can be used to compute the r -bit approximation of $(x, a_0, \dots, a_d)$.)

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- There could be many polynomials passing through the above point.

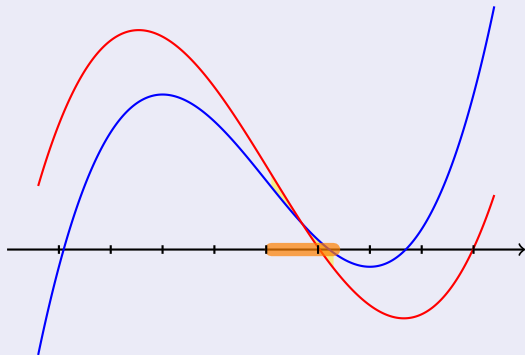
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- There could be many polynomials passing through the above point.
- What is the guaranty that every such polynomial curve has *sufficient* information?

Proof (cont...):

Main Theorem 1

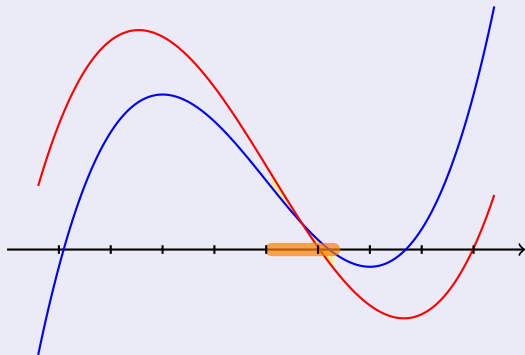
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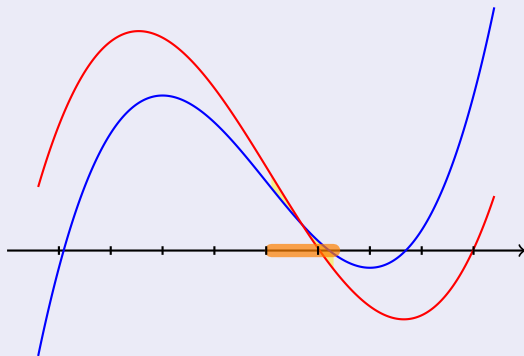
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- For two intersecting polynomials, either they are *close-by*, or one of them has high complexity (orthogonal).



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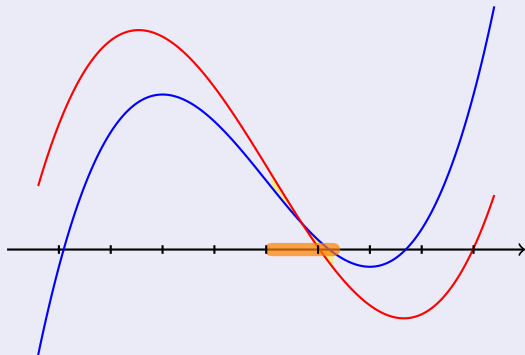
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- Unlike lines, polynomials need not have at most one point of intersection.

Key idea:

Instead of approximating the *area* of intersection, consider all *potential* points of intersection, and return a *small* list of candidates, for every precision r .



Theorem 2 (K., Nandakumar, 2026)

For a degree d polynomial $\sum_{i=0}^d a_i x^i \in \mathbb{R}[x]$ such that $\dim(a_0, \dots, a_d) \leq 1$,

$$(\forall s \in [0, 1]) (\exists x \in \mathbb{R}) \left(\dim \left(x, \sum_{i=0}^d a_i x^i \right) = s + \dim(a_0, \dots, a_d) \right).$$

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The dimension spectrum of such polynomials contains a unit length interval.

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Simple constructions like adding chunks of a random string followed by zeros, or periodical replacement of random bits with zeros don't work.
- Construct a real $x \in \mathbb{R}$ by alternating segments of a random $y \in \mathbb{R}$ and approximations of the coefficients $a_0, \dots, a_d$ of the polynomial.

Proof (cont...):

- In other words, for the j^{th} -block $x[h_{j-1}, \dots, h_j]$ of x , and an ML random $y \in \mathbb{R}$ (with respect to x),

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$$x[h_{j-1} \dots sh_j] = y[h_{j-1} \dots sh_j]$$

$$\begin{aligned} x[sh_j \dots h_j] &= a_1[0]a_2[0] \dots a_d[0]a_1[1]a_2[1] \dots a_d[1] \dots \\ &\quad \dots a_1[i_j - 2]a_2[i_j - 2] \dots a_d[i_j - 2]a_1[i_j - 1]a_2[i_j - 1] \dots a_d[i_j - 1]. \end{aligned}$$

(Note that a_0 is not being encoded in the string.)

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(Note that a_0 is not being encoded in the string.)

- For the first part of block j , we essentially have a random string.
- For the second part of the block, we use the coinciding polynomial argument from the previous result.

Proof (cont...):

- Hence, we get $K_r \left(x, \sum_{i=1}^d a_i x^i \right) \geq (s + \dim(a_0, \dots, a_d) + \gamma) r$, where γ is an error term, sub-linear in r .

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- Now, we compute the information bounds for the possible values of $s \in [0, 1]$.

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- Now, we compute the information bounds for the possible values of $s \in [0, 1]$.
 - $s = 0$.
(x contains information fetched only from the polynomial.)
 - $s = 1$.
(x is a completely random point with respect to the polynomial.)
 - $s \in (0, 1)$.
(This case can be further divided in to 2 cases based on the information content of the polynomial.)

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 - $s \in (0, 1)$.
(This case can be further divided in to 2 cases based on the information content of the polynomial.)
- For all cases, we get $\left(\dim \left(x, \sum_{i=0}^d a_i x^i \right) = s + \dim(a_0, \dots, a_d) \right)$.



- Polynomials with coefficients with high effective Hausdorff dimension.
- General algebraic curves in the Euclidean space.

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Thank you!